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Osimertinib in Untreated EGFR-Mutated Advanced Non-Small-Cell Lung Cancer (FLAURA)

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Introduction

Lung cancer is the leading cause of cancer-related mortality globally. In patients with non-small-cell lung cancer (NSCLC) harboring epidermal growth factor receptor (EGFR) mutations, EGFR tyrosine kinase inhibitors (TKIs) are the standard of care. However, resistance frequently develops. Osimertinib (Tagrisso) is a third-generation EGFR-TKI that irreversibly inhibits both EGFR-sensitizing and EGFR T790M resistance mutations.

Methods

In this double-blind, phase 3 trial, we randomly assigned 556 patients with previously untreated EGFR-mutated (Exon 19 deletion or L858R) advanced NSCLC in a 1:1 ratio to receive either osimertinib (80 mg once daily) or a standard EGFR-TKI (gefitinib or erlotinib). The primary end point was investigator-assessed progression-free survival (PFS).

Results

The median progression-free survival was significantly longer with osimertinib than with the standard EGFR-TKIs (18.9 months vs. 10.2 months; hazard ratio for disease progression or death, 0.46; 95% confidence interval [CI], 0.37 to 0.57; $P < 0.001$). The response rate was similar in the two groups (80% and 76%). The median duration of response was 17.2 months with osimertinib and 8.5 months with standard EGFR-TKIs. Overall survival at 18 months was 83% with osimertinib and 71% with standard EGFR-TKIs (hazard ratio for death, 0.63; 95% CI, 0.45 to 0.88; $P = 0.007$).

Toxicity and Safety

Adverse events of grade 3 or higher were less frequent with osimertinib (34% vs. 45%). The most common adverse events included diarrhea and dry skin. Osimertinib demonstrated a superior safety profile compared

to first-generation TKIs.

Conclusion

Osimertinib showed efficacy superior to that of standard EGFR-TKIs in the first-line treatment of EGFR-mutated advanced NSCLC, with a similar safety profile and lower rates of serious (grade 3 or higher) adverse events. It has established a new global standard of care for our patients like P001, P007, and P012.